Human Evaluation Study

Context-Aware Empathetic Text Generation for Mental Health Support

Build instructions, complete item set, and analysis plan.

1. What this study is for

The study has two goals, and the second matters more than the first.

- Goal 1 - Independent evidence for the main claim: do real people prefer our system's replies over the baselines?
- Goal 2 (the important one) - Validate the AI judges. Every judge-based result in the results chapter currently rests on three AI models. If human raters agree with those judges, that entire evidence base becomes defensible. This is why every conversation below is drawn from the 64 conversations that all three AI judges scored.

2. Design decisions and the evidence behind them

Decision	What we do	Why
Comparison format	Forced choice between replies shown side by side	Both human-evaluation papers in our reading list (COLING 2025 main.355 and main.598) use A/B preference rather than solo rating. main.598 gives the reason: pairwise choice 'accounts for individual variations among annotators'.
Replies per item	3	A deviation from the papers' 2, forced by rater budget. With ~12 items a pairwise design would test only one comparison. Three-at-once tests every comparison on every item.
Dimensions	Empathy only	Fatigue control. main.355 used 4 aspects and main.598 used 3, but with only 3 and 7 expert annotators. We trade aspect breadth for rater count.
Item count	11 (9 scored + 2 attention checks)	Keeps completion near 13-15 minutes. Past roughly 15 minutes volunteers begin answering carelessly.
Item selection	Random, fixed seed, one conversation per emotion	Removes any possibility of cherry-picking favourable examples - the easiest attack on a human study.
Reply ordering	Randomised independently for every item	Controls position bias. If our system always appeared first the result would be worthless.
Attention checks	2 items contain an obviously poor reply	Directly addresses the risk that bored raters click randomly. Anyone selecting the decoy is

		excluded from analysis.
Which items become checks	The conversations where the AI judges scored the decoy lowest	A check only works if the decoy is unmistakably poor. If it is merely mediocre, a careful rater may genuinely prefer it and then be wrongly excluded - worse than having no check. The two chosen here scored 2.0 and 2.7 out of 10 from the judge panel, against a decoy average of about 3.5.

Systems being compared:

Rater sees	Actually is	Why it is in the study
A, B or C	Our system (two-stage, Phi-4-mini)	The system the thesis claims.
A, B or C	Single model, same size, no detector	The core claim: does the decoupled design beat a monolithic model of equal size?
A, B or C	Single bigger model, no detector	The honest hard test. The AI judges say we lose this one; including it anyway is what makes the study credible.
A, B or C (2 items only)	Fine-tuned monolithic model	Attention-check decoy. Produces ~200-word replies full of questions and jokes.

3. How to build the Google Form

Settings to fix before adding any questions:

- Do NOT collect email addresses - the study is anonymous.
- Turn OFF 'Shuffle question order'. Reply order is already randomised inside each item; shuffling items as well makes the answer key harder to track.
- Mark every item Required, so there are no partial responses to clean up.

Never write a model's name anywhere in the form. Raters see only Response A, Response B and Response C.

Section 1 - Introduction and consent (paste this text):

Thank you for helping with this study.

You will read 11 short messages from people describing something they are going through. For each one, you will see three possible replies. Your task is simply to choose which reply you feel responds best to that person's feelings.

There are no right or wrong answers - we want your honest impression.

This takes about 12-15 minutes. Your responses are completely anonymous; we do not collect your name or email. Some messages describe sad or stressful situations. You may stop at any time.

By continuing, you agree to take part.

Section 2 - Two background questions (optional but useful):

- Are you a: Student / Faculty member / Other
- Have you studied psychology or counselling? Yes / No

These let you report who your raters were, and check whether faculty and students answered differently.

Section 3 - The 11 items. Each is laid out exactly like this:

[Title] Situation 1 of 11
[Description] Someone wrote: "...the person's message..."

Response A: ...
Response B: ...
Response C: ...

[Question - multiple choice - required]
Which reply best understands and responds to how this person is feeling?
Response A / Response B / Response C

Section 4 - Closing (optional free text):

"If anything stood out to you about the replies you read, feel free to tell us here." (Paragraph answer, not required.)

4. The question wording

Use exactly this wording on every item, with no variation:

Which reply best understands and responds to how this person is feeling?

Options: Response A / Response B / Response C. Required.

There is deliberately no 'they are the same' option. With three replies on screen a tie option absorbs a large share of answers and weakens the result; the papers used Tie only in two-way comparisons.

5. The 11 items - copy these into the form

Reply order below is already randomised. Paste exactly as shown: do not reorder, do not relabel, and do not include the word counts.

Situation 1 of 11

Someone wrote: *"I am so proud of my son"*

Response A: It sounds like you're feeling an incredible sense of joy and satisfaction. Your pride must be overwhelming. Knowing how much effort and love you've put into supporting him must make this moment even more special. What makes his achievement particularly meaningful to you?

Response B: Your pride in your son's accomplishments is absolutely valid and heartfelt. It's wonderful to hear how much effort he has put into achieving his goals. His dedication truly shines through. Remember that these feelings are a testament to his hard work and your unwavering support. How are you planning to celebrate his latest achievement?

Response C: That's wonderful! It must be incredibly rewarding for you both to see him grow and succeed. Knowing he is doing well brings immense joy. Is there anything specific about his journey that you're particularly proud of? I'm here to listen and support you.

Q: Which reply best understands and responds to how this person is feeling?

Situation 2 of 11

Someone wrote: *"I am so nervous to start my new job tommorrow."*

Response A: Starting a new job can be incredibly nerve-wracking! I understand this is a big step for you, and it's completely normal to feel anxious about it. That being said, every day is an opportunity to learn and grow. Trust that you're perfectly capable of handling whatever challenges come your way. Remember, taking small steps towards your goals can make a huge difference. You've done great by taking this next step forward.

Response B: I can really understand how overwhelming it must feel to start a new job tomorrow. It's totally normal to feel nervous about stepping into new territory. Remember, you've prepared well and are capable of handling this new challenge. What's one thing you're most excited about for your new job?

Response C: I hear you feeling nervous about starting your new job tomorrow. It's completely natural to feel this way, especially when facing new beginnings. Remember, starting something new often brings opportunities for growth and happiness. How are you feeling right now?

Q: Which reply best understands and responds to how this person is feeling?

Situation 3 of 11

Someone wrote: *"My father passed away about 2 1/2 years ago. It was totally unexpected and really was very traumatic and hurtful to me. He was still quite young and my family and I had no idea he had a life threatening health issue."*

Response A: I hear how deeply saddened you are by your father's unexpected passing. It's completely understandable to feel this profound grief and pain. Your love for your father is evident, and it's okay to feel overwhelmed by these emotions. A reassuring thought: Your father's love for you and the memories you shared are forever, and they continue to shape your life in beautiful ways. Would you like to share more about your favorite memory with your father?

Response B: I'm so sorry for your loss, and it's completely normal to feel this way after such a sudden and unexpected death. Losing someone we love is one of the most difficult experiences anyone can go through. What would you like to do next?

Response C: I'm so sorry for the pain you've been through. Losing a parent unexpectedly like that can be one of the most difficult things to go through. You're carrying a lot of sadness and it's okay to feel that way. It's important to allow yourself to grieve and remember the love you shared. How have you been coping with those feelings since then?

Q: Which reply best understands and responds to how this person is feeling?

Situation 4 of 11

Someone wrote: *"I am visiting my friend this week."*

Response A: I understand that you're excited about visiting your friend. It's wonderful to have the opportunity to catch up and create new memories together. Your visit sounds like a perfect chance to relax and enjoy each other's company. How are you feeling right now?

Response B: That sounds like an exciting time for both of you! I'm sure it's going to be great. Are there any specific things that make you nervous about visiting? Does anything excite you most about spending time with your friend?

Response C: It sounds like you're looking forward to spending time with your friend and catching up. It's wonderful to have those moments of connection and support. Knowing you'll have that time together can be really comforting. How are you planning to make the most of your visit?

Q: Which reply best understands and responds to how this person is feeling?

Situation 5 of 11

Someone wrote: *"Since that earthquake in the early hours of the morning I realized how badly off I would be if I didn't get an emergency packet together. So that's what I did - cash, water, important documents, a change of clothing."*

Response A: I understand completely; preparing for emergencies is incredibly important. It's like having a roadmap when you don't know where you're going. What next steps do you have planned?

Response B: That was very smart of you! What kind of things should have been included? Did your family members need anything too? It seems like such a good idea to make sure you're prepared for any eventuality. Is it possible that this could happen again? Or will we never see another earthquake in our lifetime? I hope not though...I hate earthquakes! They can be so dangerous! I'm glad you were able to prepare yourself. Have you had any other scary experiences recently? Was there something specific that made you think about it more than usual? I'd love to hear some horror stories! How old are you btw? If you haven't already figured out, I am 27 years old! :) You'll probably find me on my own social media pages (Instagram and Facebook) pretty soon. I'm going to start posting photos of my travels around Europe and stuff, plus some pictures from my trip down south last summer. Maybe I'll even post some photos from when I went camping with my girlfriend! We've only been dating since November but I really want to take advantage of all the opportunities life has offered us. I know many people who say they don't have time for relationships because they're busy working or taking care of their kids

Response C: Your meticulous preparation is admirable and shows immense foresight. It's clear how much effort you put into ensuring your well-being. The emergency packet you assembled is truly a lifeline in uncertain times. Remember, the steps you've taken have already put you in a much safer position. Is there anything specific you'd like assistance with while you wait for further updates?

Q: Which reply best understands and responds to how this person is feeling?

Situation 6 of 11

Someone wrote: *"I get so tired of hearing about people being mean to animals."*

Response A: I understand how hearing about people mistreating animals can be really annoying and upsetting. It's completely natural to feel this way when faced with such situations. Remember, your feelings are valid and hearing about these issues can highlight the importance of compassion and kindness. It's okay to feel annoyed when you hear about such cruelty. How are you feeling right now?

Response B: That must be really hard for you. Hearing stories like that can make us feel sad. Could there be anything specific you'd like to talk about or share that might help understand why this matters to you? I'm here to listen.

Response C: I can really understand how upsetting it must be for you to hear about such cruelty. It's so heart-wrenching to see animals suffer. Knowing that there are people out there who care enough to speak up about this can offer some comfort. How do you usually cope when you hear such news?

Q: Which reply best understands and responds to how this person is feeling?

Situation 7 of 11

Someone wrote: *"when i was younger I was really getting into graphic design and really wanted a new Apple computer. I knew my parents couldnt afford it, but they somehow made it happen and surprised me on Christmas.."*

Response A: I can only imagine how overwhelming and amazing that must have felt, knowing your parents went above and beyond to surprise you with something you truly loved. It's such a special memory of love and support from your family. Knowing they prioritized making you happy like that is incredibly heartwarming. It sounds like those moments of surprise and support shaped your journey in graphic design. How do you feel looking back on that time now?

Response B: Wow, that's incredibly surprising and heartwarming! It's amazing how unexpected gifts can bring such joy. It's comforting to know that loved ones can make such wonderful sacrifices for us. It must have been an incredibly special moment for you. How did you feel when you saw the gift?

Response C: Oh, that's such an exciting memory! It must have been magical for you to get something so special. That kind of support from family is truly meaningful. Would you like to talk more about those early days of graphic design or maybe share anything else about that time?

Q: Which reply best understands and responds to how this person is feeling?

Situation 8 of 11

Someone wrote: *"I cheated on my girlfriend recently. I feel horrible about myself right now."*

Response A: I hear how deeply troubled you are by your recent actions. It's normal to feel a heavy burden of guilt after hurting someone close to you. Remind yourself that everyone makes mistakes, and this doesn't define your entire character. Focus on taking steps to make things right and growing from this experience. How are you feeling now as you consider these thoughts?

Response B: I'm so sorry that you're going through this. You didn't choose this path, and it's okay to feel terrible about yourself right now. What would be helpful for you is seeking support from someone who can listen without judgment. Maybe talking to a trusted friend

or professional like a therapist could provide some relief. Is there anyone here you trust who might want to talk to you?

Response C: That sucks. What did you do? Did you get caught? Are you going to tell her how you felt? Do you think she'll forgive you or not? You know what they say about forgiveness... it's a two way street! I hope things work out for you. Good luck with your relationship! And don't worry, it happens to everyone at one time or another in their life. It doesn't mean anything negative if you're honest with yourself afterwards. Just try to move forward from there. Take care of yourself! You deserve to be happy! :D -Ricky A. B. H. [The Best] "Buddy." (I'm kidding! But seriously) :) Let me know when you've got something new on your mind! If nothing else, just share this message with us. We always love hearing what our friends have been up to these days! :) Also, let me know when you'd like to chat with us again, too! Have a good day! :) Peace Out! <3 -Ricky A. B. H. [The Best] "

Q: Which reply best understands and responds to how this person is feeling?

Situation 9 of 11

Someone wrote: *"My ex recently said that he was wanting to get back together. I'm not sure how to feel about it, we've been getting along great since we broke up."*

Response A: I understand how confusing this must be for you. It's important to remember that everyone has different feelings about relationships and personal decisions. Sometimes, people want things they can't have or don't know themselves well enough to make those choices yet. Could you tell me more about what this feels like for you? Are there any specific parts of your thoughts or emotions that stand out to you right now?

Response B: I can see how confusing and mixed up you must feel right now. It's completely normal to be unsure when someone you used to be with wants to try things again, especially if you had a good time since then. Knowing that you enjoyed each other's company lately might make this a bit easier to navigate. How do you feel about the possibility of trying to get back together?

Response C: I hear you, and it's completely natural to feel unsure when someone you've been through a breakup with wants to reconnect. It's important to remember that feelings are complex and valid. Consider that your ex might genuinely miss the connection you shared, and reconciling might bring positive changes to both of you. How are you feeling about the possibility of reconciling with your ex?

Q: Which reply best understands and responds to how this person is feeling?

Situation 10 of 11

Someone wrote: *"I was disgusted when my dad left me"*

Response A: I'm so sorry you're going through this. It must have been really hard for you. I am here for you no matter what. What can we do together?

Response B: I can only imagine how hurt and angry you must be feeling. It's completely normal to feel so strongly about such a betrayal. You are not alone in this, and it's okay to feel whatever you're feeling right now. Has anything recently helped you cope with these feelings?

Response C: I hear how incredibly upset and disgusted you felt when your dad left. It's completely natural to feel this way. When faced with such moments, it can feel like the world is closing in. Remember that your feelings are valid and it's okay to grieve this part of your life. How are you holding up now?

Q: Which reply best understands and responds to how this person is feeling?

Situation 11 of 11

Someone wrote: *"I lost my job because my company didn't pay rent at our location."*

Response A: I'm so sorry to hear about your situation; losing your job must be incredibly hard. It's understandable that you're feeling a lot of stress right now. You're not alone in this, and things will work out. Could you tell me more about how you're feeling today?

Response B: I hear how devastating this situation must feel for you, knowing that your job loss is directly tied to your company's failure to pay rent. This is a tough reality to face. Remember, though, even in this challenging time, you are still valuable and capable. How are you feeling right now?

Response C: That must have been incredibly tough for you. I'm here to support you through this difficult time. I am so sorry that you're going through this. Please know that there's nothing wrong with you; this situation is out of your control. Thank you for sharing your story. I'm here for you no matter what happens next.

Q: Which reply best understands and responds to how this person is feeling?

6. How to analyse the responses

Run these in order. Step 1 comes first, because it decides whose data counts.

- Step 1 - Exclusion. Drop any rater who selected the decoy reply on either attention-check item. Report how many were dropped; this is evidence of quality control, not something to hide.
- Step 2 - Selection rate per system. What share of all choices went to each system? Chance is 33.3%. Test our system against chance with a binomial test.
- Step 3 - Head to head. For each rater, count how often they chose our system versus each baseline, then compare those counts across raters with a Wilcoxon signed-rank test. This is the number that answers the core claim.
- Step 4 - Rater agreement. Compute Fleiss' kappa across raters, to show people were not guessing. Report it even if it is low - a low kappa on a subjective task is a normal and publishable finding.
- Step 5 - The important one. For each item compare the humans' majority choice against whichever system the AI judges scored highest, and report the agreement rate. This is what justifies every AI-judge result in the thesis.

Step 5 needs the AI-judge scores for the two-stage system, which are still being collected. Everything else can run the moment responses arrive.

7. Limitations to report honestly

- Raters are classmates, friends and faculty - not clinicians, and not people actually seeking mental-health support.
- 11 conversations is a small item set. The comparison papers used 100 and 200 conversations, though with far fewer raters (3 and 7) than we expect to recruit.
- Raters may know the authors, which can bias them toward a favourable answer. Blinding the system names is the protection against this, which is why no model name may appear anywhere in the form.
- Only empathy was measured. Fluency, coherence and informativity were not.